

Balloon Sat Arduino Shield Project

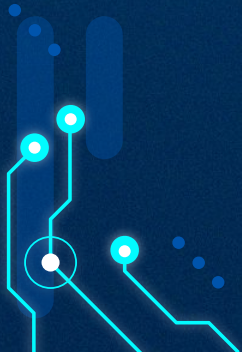
Michael Kurta

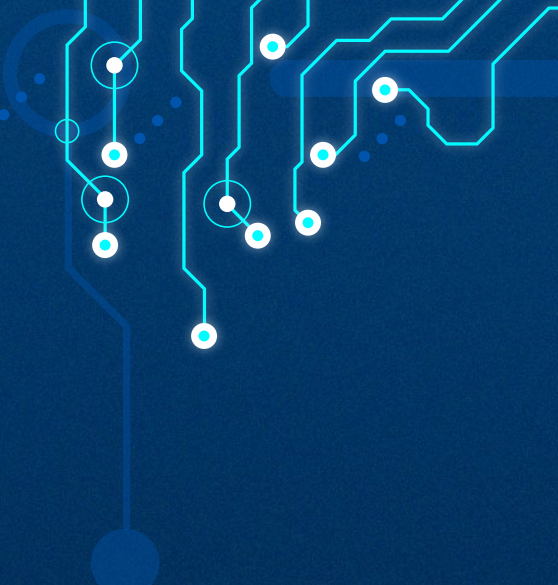


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A more thorough approach



An abstract circuit pattern in the top-left corner, featuring glowing cyan lines and nodes on a dark blue background.

01

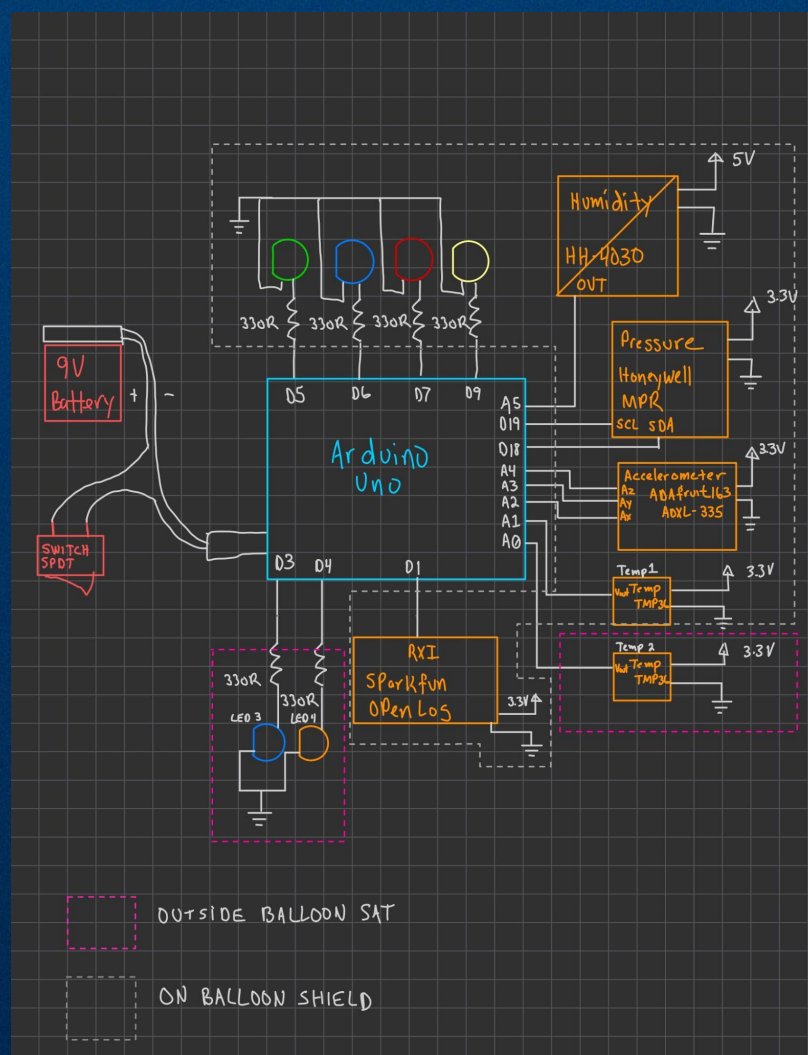
PHASE 1

An abstract circuit pattern in the bottom-right corner, featuring glowing cyan lines and nodes on a dark blue background.

Original Design – BOM

- 1x Arduino Uno
- 1x 9V battery
- 1x 9V battery to Arduino Barrel Jack
- 1x PCB to function as a shield for Arduino
- 1x Sparkfun OpenLog module (UART)
- 1x Honeywell HH-4030 Humidity Sensor Module (No longer in production)
- 1x Adafruit ADXL 335 Accelerometer Module
- 1x Sparkfun Qwiic Micropressure Sensor
- 2x Analog Devices TMP36 Temperature Sensors
- 2x Blue 5mm LEDs
- 1x Orange 5mm LED
- 1x Yellow 5mm LED
- 1x Green 5mm LED
- 1x Red 5mm LED
- 2x 8-Pin stackable plug headers
- 2x 6-Pin stackable plug headers
- 2x DIP socket halves
- 2x 6-Pin socket headers
- 1x 3-Pin socket header
- 2x 2-Pin plug breakaway headers
- 1x 3-Pin locking headers
- 1x 6-Pin breakaway header
- 1x 3-Pin plug breakaway header

Original Design - Block Diagram



Cost ~\$110



02

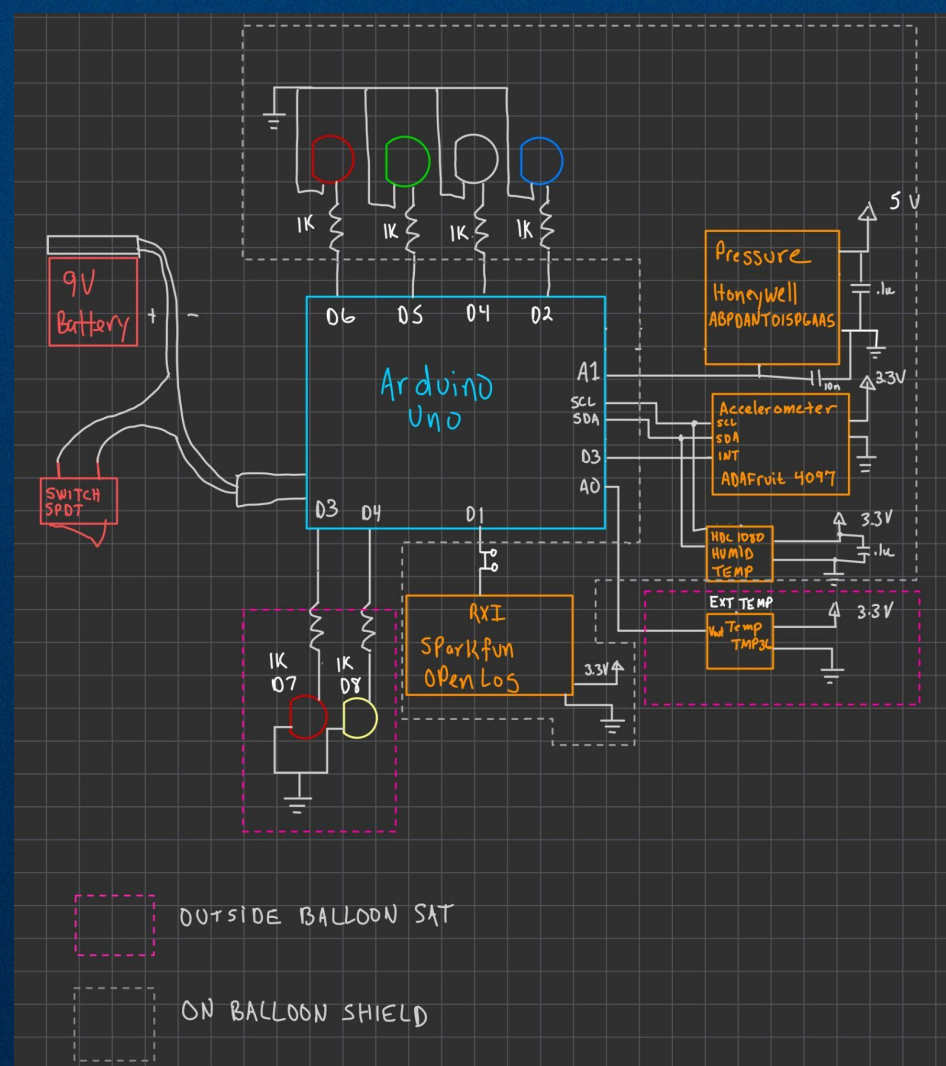
PHASE 2

PHASE 2 – BOM

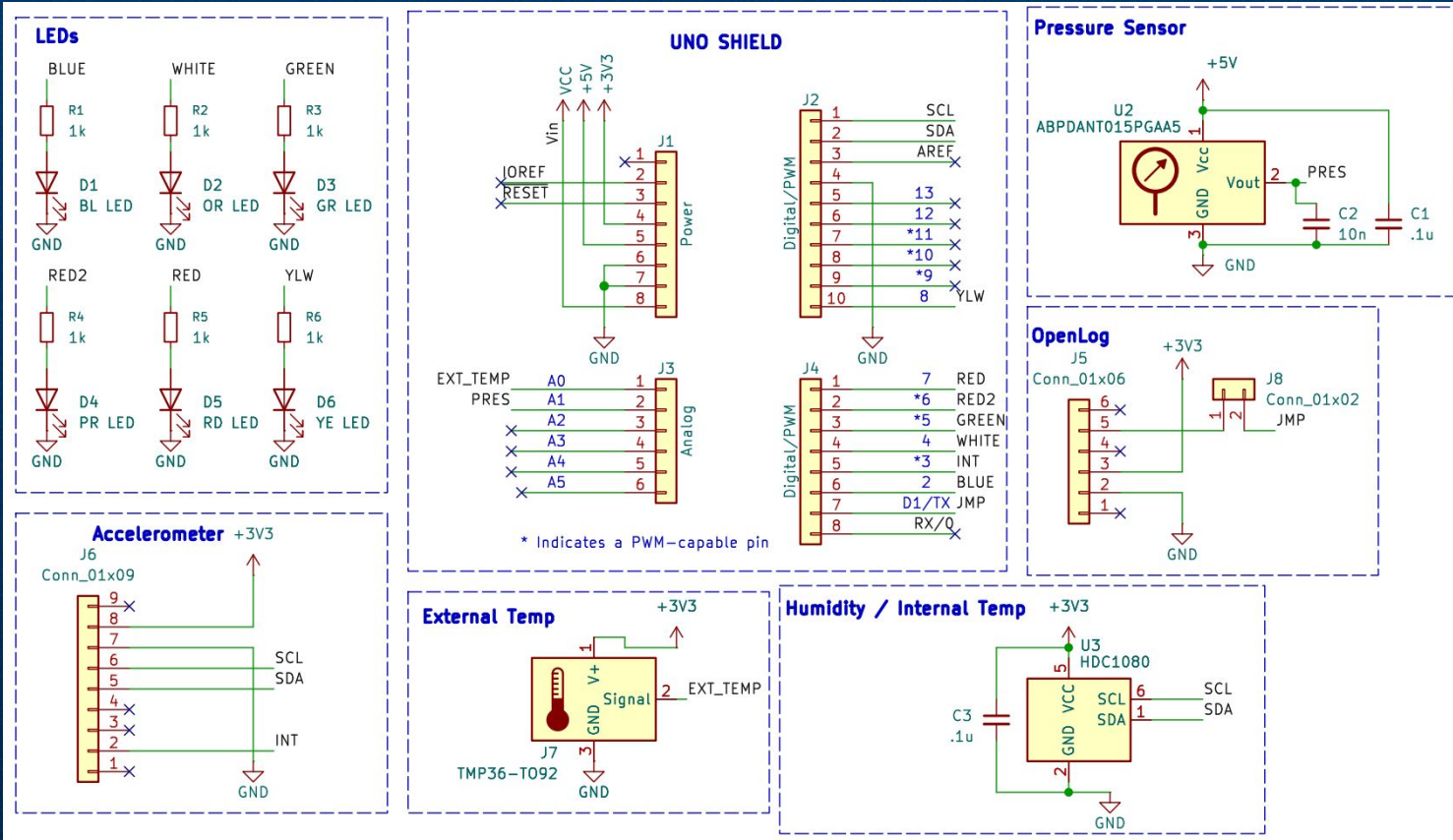
Part	Name	Link	Price w/o shipping (total)	Avail?
Open Log	Sparkfun Open Log		\$21.50	Yes
Accelerometer	ADXL343 - TRIPLE-AXIS ACCELEROMETER		\$5.95	Yes
Temp Sensor			\$1.78	Yes
Rocker switch			\$5.72	Yes
PCB	2.6x2.1 in Osh Park After Dark 2-layer		\$30.00	Yes
Humidity / Temp			\$2.19	Yes
Decoupling cap	1x 10nF 25V		\$0.54	Yes
Decoupling cap	2x 100nF 50V		\$1.74	Yes
5 color 5mm LED kit			\$4.95	Yes
pressure 2			\$28.10	Yes
JST-PH 3 pin horizontal female			\$3.50	Yes
JST-PH 3 pin male			\$1.25	Yes
Battery Connector			\$2.00	Yes
Heat shrink tubing	1/16"		\$4.00	Yes
32GB SD Card			\$9.42	Yes
Silicone wire 22AWG 25'			\$8.27	Yes
1k Ohm resistors			\$0.60	Yes
			\$131.51	

[illegible]

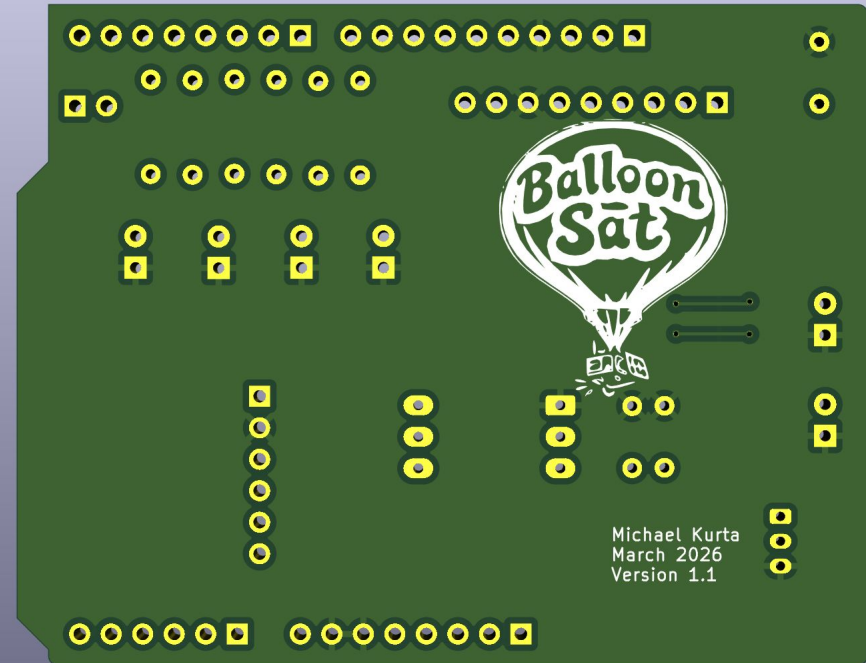
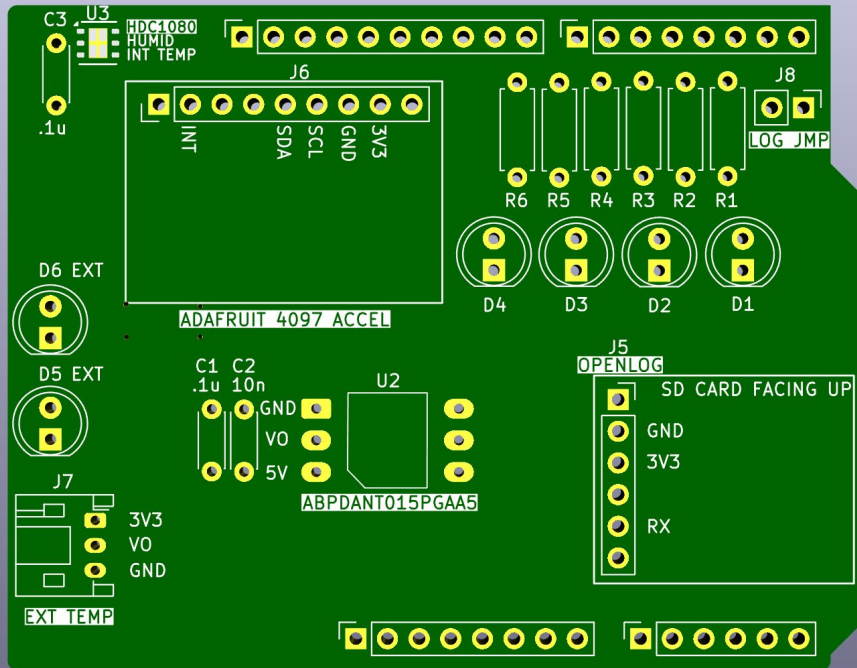
PHASE 2 - Block Diagram



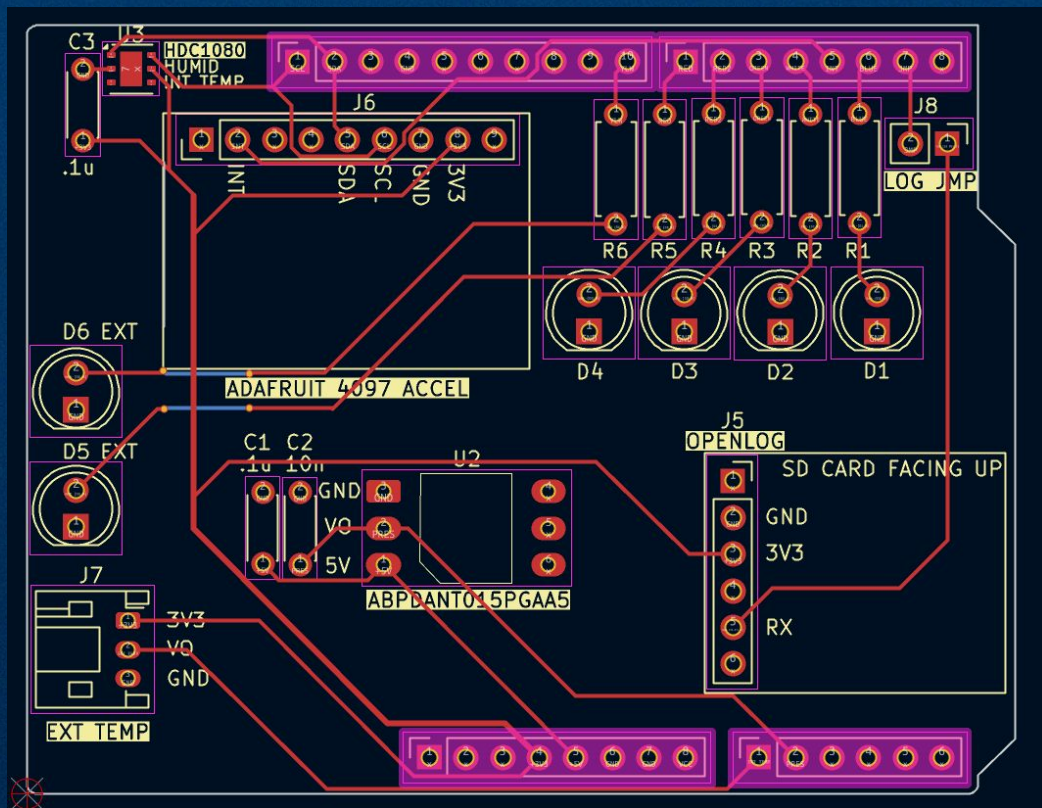
PHASE 2 - Schematic



PHASE 2 – PCB

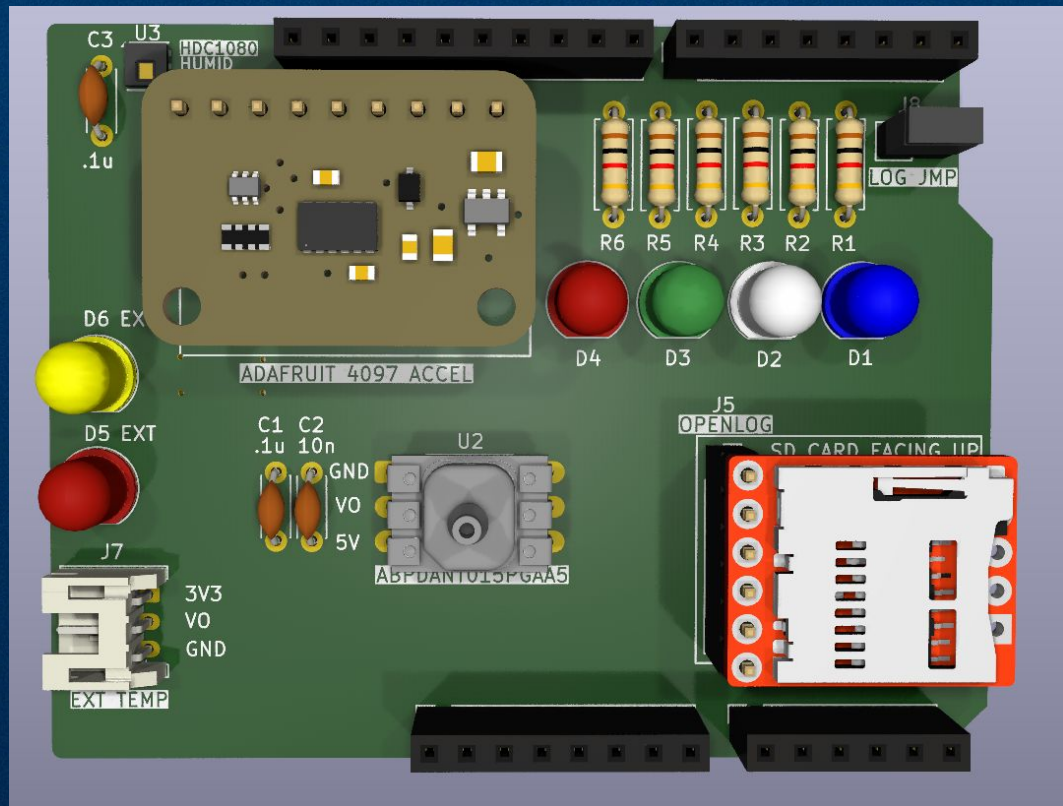


PHASE 2 – PCB



PHASE 2 – PCB & Features

- Connector for external Temp sensor
- Decoupling capacitors
- No additional I2C resistors
- Easy access SD card
- All external wiring in one area on the board
- Jumper to prevent bootload writes to SD card
- Dual-use temp / humidity sensor
- Part orientation easily marked
- Accelerometer INT pin connected to Arduino interrupt trigger pin



THANKS!

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